

Author Contribution Statement

Using the CRediT (Contributor Roles Taxonomy) categories. This draft reflects the project's evident structure – a candidate/first-author driving the implementation and experiments under a named supervisor, per the TCG CREST/CQuERE ARC program record in `latex_outputs/paper_latex/hvk_com` cover page – but the specific role assignments below are a starting draft, not a verified record: only the authors know the true division of labor, and this should be corrected before submission if it misrepresents either author's actual contribution.

Sparsho Chakraborty: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization.

Dr. Siddhartha Patra: Conceptualization, Supervision, Writing – review & editing, Resources.

[FILL: confirm/adjust – e.g. did the supervisor also contribute Funding acquisition, Project administration, or direct Methodology input?]

[FILL: If any additional collaborators, hardware-access sponsors, or dataset providers materially contributed but are not listed as authors, add an Acknowledgments section here or in the manuscript – do not add them as authors without their consent.]